

ENV 6106

Air Pollution Modeling

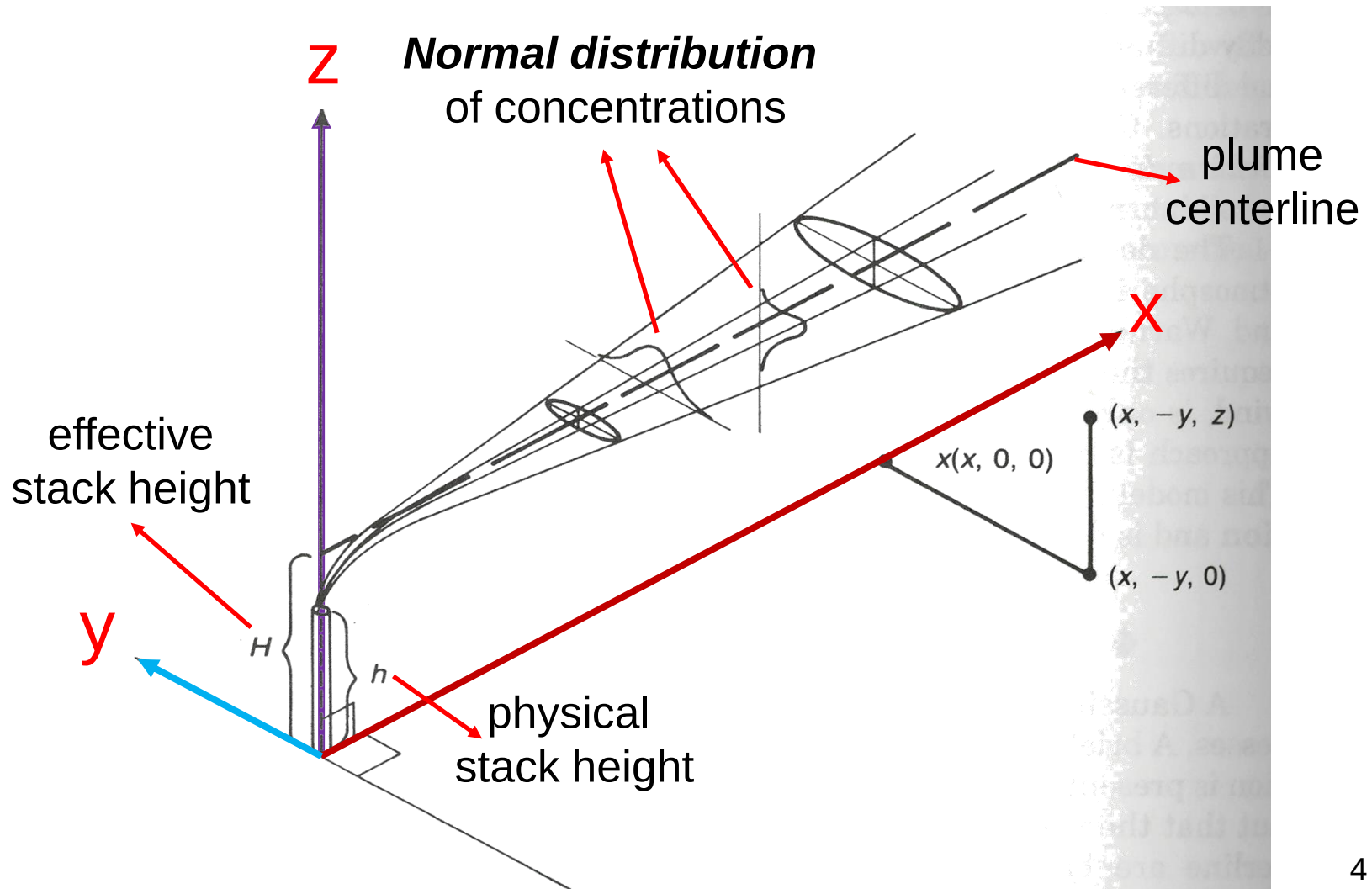
Spring 2023

Topics

- Gaussian model, plume rise and downwash
- Expectation
 - Be able to interpret how ground concentration varies using Gaussian plume model
 - Understand common models for plume rise
 - Understand downwash
- Readings: Cooper & Alley Chapter 20.1-20.4

More on Gaussian model

Inside the time-average plume...



The Gaussian Plume Model

- Take account into ground reflection

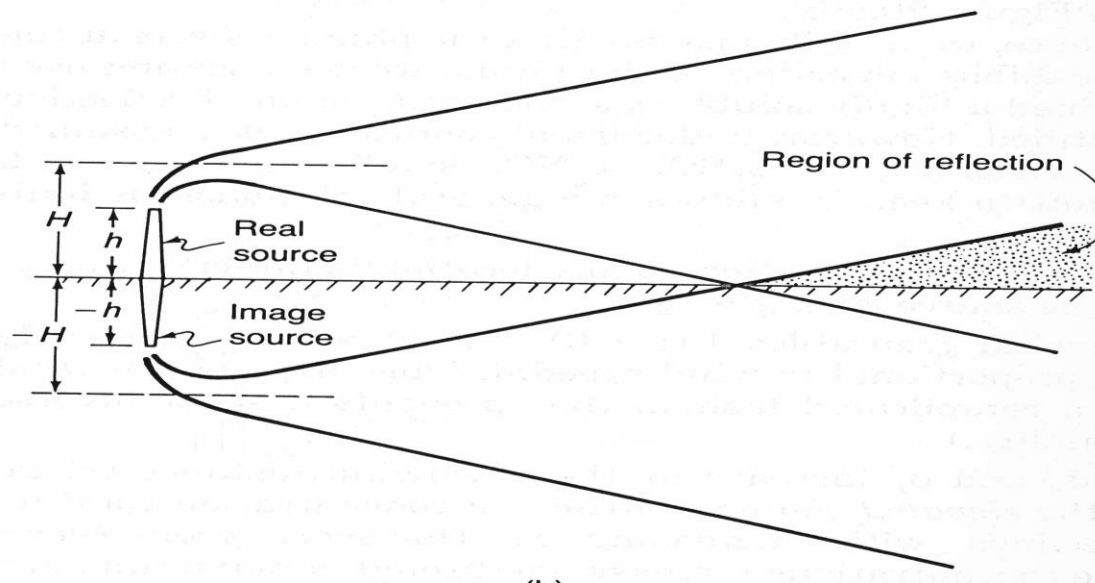


Figure 20.5 (Cooper and Alley, 2002)

$$C(x,y,z) = \frac{S}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left\{ \exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right) \right\}$$

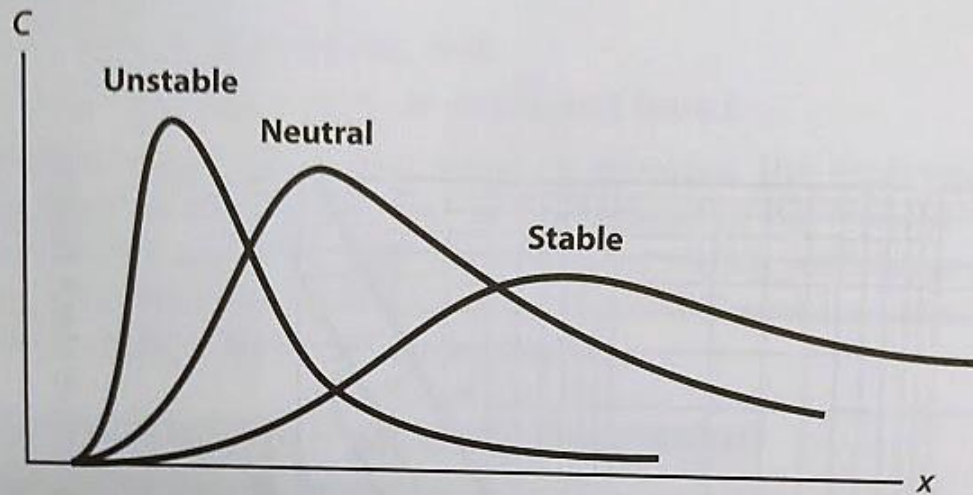
*This one is the **most used** form of Gaussian plume model*

The Gaussian Plume Model

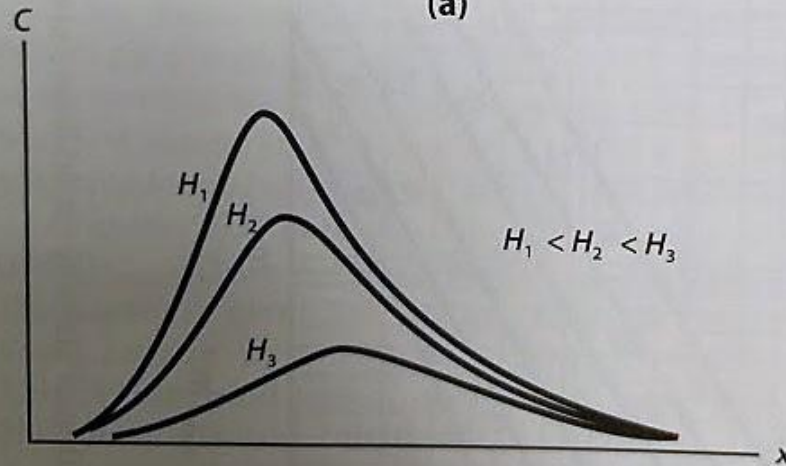
- C proportional with source strength S
- Ground C inversely proportional with stack height H
- Along X direction, C starts at 0, increase to maximum concentration, then decreases again. The distance of max concentration increases with stack height H
- Stack height design: high enough so ground level concentration not exceed certain threshold, but just high enough so no need to spend un-necessary money

$$C(x,y,z) = \frac{S}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left\{ \exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right) \right\}$$

Ground concentrations



(a)



(b)

Figure 20.10

Behavior of ground-level centerline concentration from an elevated source as a function of downwind distance and (a) atmospheric stability (at constant H), and (b) effective stack height (at constant stability).

(Cooper and Alley, 2002)

Max ground concentrations

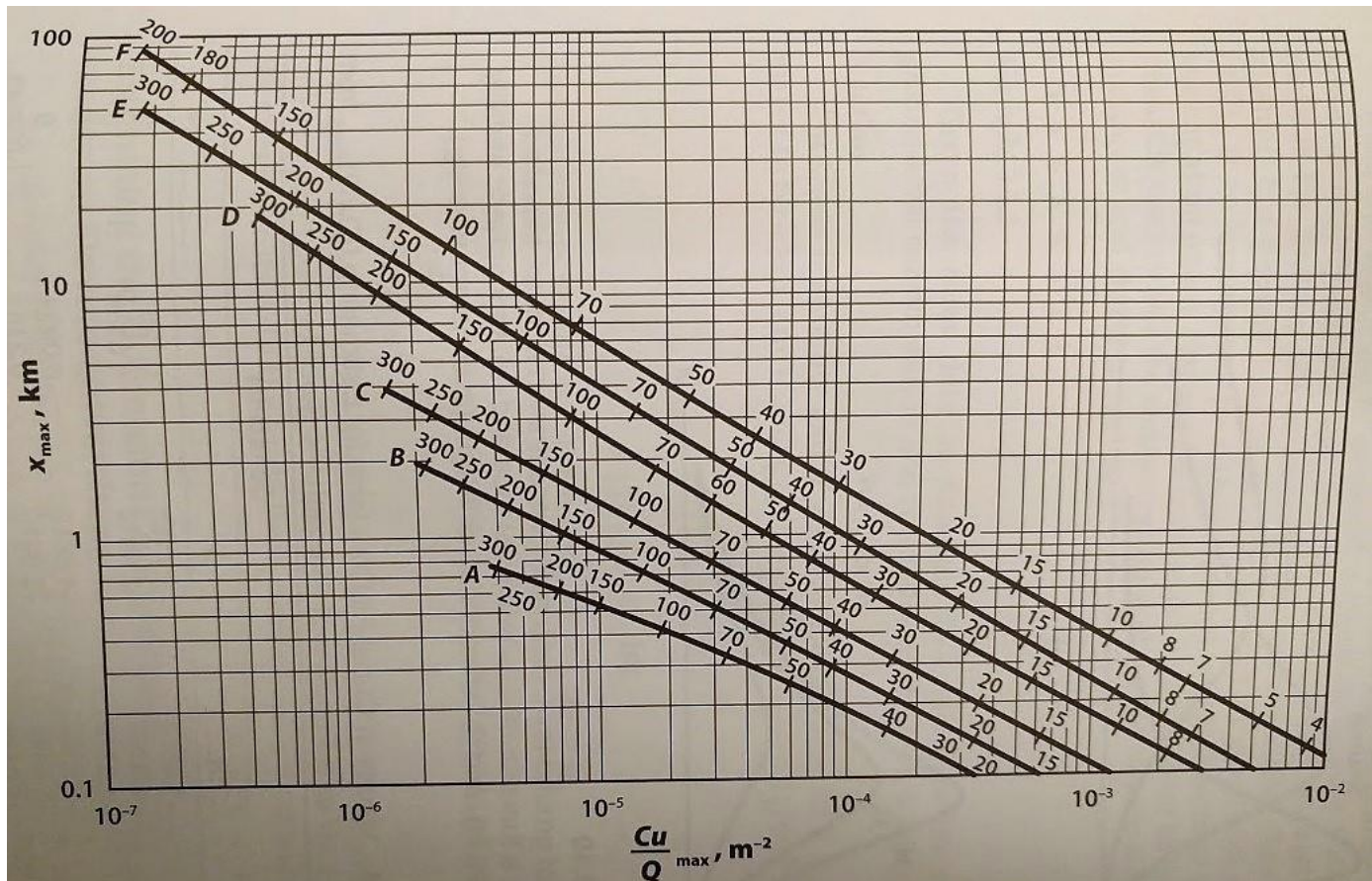


Figure 20.9

Maximum (Cu/Q) and distance to $C_{\max}$ as a function of stability (solid lines) and effective stack height (numbers [in meters]). (Adapted from Turner, 1970.)

Note: S, instead of Q, was used on the slides to represent emission rates
(Cooper and Alley, 2002)

Max ground concentrations

$$(Cu/S)_{max} = \exp[a + b(\ln H) + c(\ln H)^2 + d(\ln H)^3]$$

Table 20.4 Values of Curve-Fit Constants for Estimating $(Cu/Q)_{max}$ from H as a Function of Atmospheric Stability

Stability	Constants			
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>
A	-1.0563	-2.7153	0.1261	0
B	-1.8060	-2.1912	0.0389	0
C	-1.9748	-1.9980	0	0
D	-2.5302	-1.5610	-0.0934	0
E	-1.4496	-2.5910	0.2181	-0.0343
F	-1.0488	-3.2252	0.4977	-0.0765

Adapted from Ranchoux, 1976.

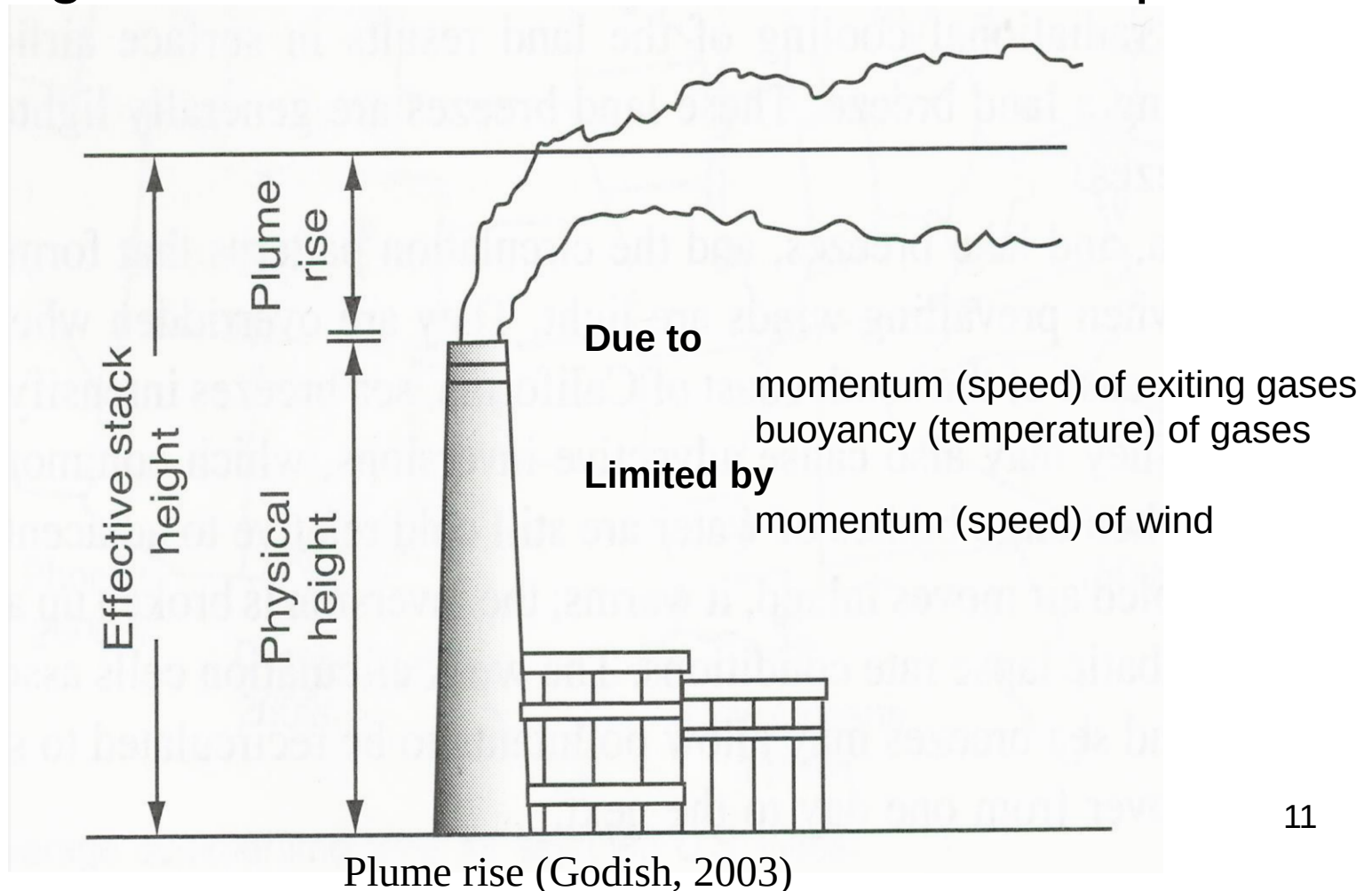
(Cooper and Alley, 2002)

Note: S, instead of Q, was used on the slides to represent emission rates

Plume rise

Plume rise

- Effective stack height (H) = physical + rise
- The figure below shows a hot and bended plume



Plume rise

- Plume rise is critical for Gaussian dispersion models
- Many formulas were proposed, provide a wide range of results, sometime up to 10x difference
- We will briefly describe a few of them
 - Holland & modified Holland, CONCAWE and Briggs
 - CONCAWE is a Dutch organization focused on oil industry
- Mostly used is the Briggs plume rise model
 - Gary Briggs, first published in 1965, updated 1969,1970,1972
 - Empirically derived based on observational data
 - Proposed for hot, bend-over plumes, less accurate for cold and wet plumes (e.g. those coming out of wet scrubber)
 - Here we discuss them in unstable and stable conditions

Height of plume rise (Δh)

- Holland

$$\Delta h = \frac{v_s d_s}{u} \left[1.5 + 2.68(10)^{-3} P_a \left(\frac{T_s - T_a}{T_s} \right) d_s \right]$$

- Modified Holland

$$\Delta h = 1.5 \frac{v_s d_s}{u} + \frac{9.6 Q_H}{u}$$

- CONCAWE

$$\Delta h = \frac{101.2(Q_H)^{0.444}}{u^{0.694}}$$

v_s = vel of stack gases, m/s

d_s = stack diameter at stack tip, m

u = wind speed at stack tip, m

T_a, T_s = abs Temp of air at stack tip, Temp of stack gases, K

P_a = absolute pressure at stack tip, mb

Q_H = heat emissions rate, MW

Briggs plume rise model

- Neutral or unstable conditions
 - 1. Buoyancy flux term

$$F_B = g \left(1 - \frac{MW_s}{28.9} \right) \left(\frac{T_a}{T_s} \right) \frac{v_s d_s^2}{4} + 8.9 \left(\frac{P_0}{P_a} \right) Q_H$$

- 2. Distance to final plume rise (x dir) and height

$$x_f = 119 (F_B)^{0.4} \quad \text{for } F_B \geq 55 \text{ m}^4/\text{s}^3$$

$$x_f = 49 (F_B)^{0.625} \quad \text{for } F_B < 55 \text{ m}^4/\text{s}^3$$

- 3. Height of plume rise

$$\Delta h = 1.6 \frac{(F_B)^{1/3}}{u} (x_f)^{2/3} \quad \text{for } x \geq x_f \qquad \Delta h = 1.6 \frac{(F_B)^{1/3}}{u} (x)^{2/3} \quad \text{for } x < x_f$$

F_B = buoyancy flux, m^4/s^3 ;

MW_s = mol wt of stack gases;

d_s = stack diameter at stack tip, m;

P_o, P_a = absolute pressure at sea level, at stack tip, mb;

T_a, T_s = abs Temp of air at stack tip, Temp of stack gases, K

x = downwind distance, m;

Δh = plume rise, m

g = gravitational constant, 9.8 m/s^2

v_s = vel of stack gases, m/s;

Q_H = heat emissions rate, MW;

u = wind speed at stack tip, m/s

Briggs plume rise model

- Stable conditions
 - 1. Buoyancy flux term

$$F_B = g \left(1 - \frac{MW_s}{28.9} \right) \left(\frac{T_a}{T_s} \right) \frac{v_s d_s^2}{4} + 8.9 \left(\frac{P_0}{P_a} \right) Q_H$$

- 2. Stability parameter

$$S = \frac{g}{T_a} \left(\frac{\Delta\theta}{\Delta z} \right) \quad \theta = T \left(\frac{P_0}{P_a} \right)^{R/c_p}$$

If data not avail, use 0.02 K/m for Class E, and 0.035 K/m for Class F

- 3. Height of plume rise

$$\Delta h = 2.6 \left(\frac{F_B}{uS} \right)^{1/3} \text{ for } u \geq 1.5 \text{ m/s} \quad \Delta h = 5F_B^{0.25} S^{-0.375} \text{ for } u < 1.5 \text{ m/s}$$

F_B = buoyancy flux, m^4/s^3 ;

MW_s = mol wt of stack gases;

d_s = stack diameter at stack tip, m;

P_0, P_a = absolute pressure at sea level, at stack tip, mb;

T_a, T_s = abs Temp of air at stack tip, Temp of stack gases, K

x = downwind distance, m;

Δh = plume rise, m;

g = gravitational constant, 9.8 m/s^2

v_s = vel of stack gases, m/s;

Q_H = heat emissions rate, MW;

u = wind speed at stack tip, m/s

$\Delta\theta/\Delta z$: potential T gradient, K/m

Max ground concentrations

- Max ground concentration does not always occur at minimum wind speed
 - Yes, C inversely associated with u
 - But, plume rise is also inversely associated with u
- The wind speed that results in max ground concentration is called critical wind speed

$$C(x, y, z) = \frac{S}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left\{ \exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right) \right\}$$

Downwash

Downwash

- As wind blows past blunt objects, flow disturbances occur, and low pressure zones are created. Depending on wind speed, vortexes and eddies are formed. These eddies can trap stack releases and bring them quickly to the ground.

Example 1: <https://www.youtube.com/watch?v=o3wPbPgQBhA>

Example 2: <https://www.youtube.com/watch?v=OqRUtRytanI>

Downwash

- Most dispersion models has taken account the effect of downwash
- Many models, such as AERMOD, uses PRIME (plume rise model enhancements). More information see Schulman et al 2000
 - <https://doi.org/10.1080/10473289.2000.10464017>
- Ways to reduce downwash:
 - Increase exit velocity ← smaller diameter
 - 1.5 times expected maximum stack tip velocity
 - Engineering method, turning wind